HEO validation benchmark for Orekit, Nyx, and SatKit

Benchmark objective

The benchmark should answer a concrete engineering question: **which toolkit provides sufficient accuracy, runtime, numerical robustness, and operational completeness for a defined highly eccentric Earth-orbit workflow?** It should not produce a single universal winner. The result should identify the best toolkit for long-arc prediction, perigee-resolving propagation, orbit determination, covariance propagation, and repeated mission-design evaluations.

Orekit is the reference feature-rich baseline because its retrieved documentation exposes analytical, numerical, and semi-analytical propagation; gravity, drag, third-body, radiation-pressure, maneuver, event, covariance, and orbit-determination capabilities.[1] Nyx should be evaluated as a performance-oriented Rust/Python alternative, while SatKit should be evaluated as a Python numerical-propagation option.[2, 3]

Test articles and reproducibility

Use one pinned release or commit of each toolkit, one compiler/interpreter environment per toolkit, and one common machine. Record CPU model, operating system, compiler flags, BLAS/LAPACK configuration, thread count, and whether acceleration hardware is used. Run each case at least 10 times after one warm-up run. Report median, interquartile range, and maximum wall-clock time rather than a single timing.

All toolkits should consume the same CCSDS-compatible initial state, epoch, frame definition, Earth-orientation data, gravity field, atmospheric model, solar and lunar ephemerides, space-weather inputs, spacecraft area-to-mass ratio, reflectivity coefficient, and maneuver sequence. If a toolkit cannot ingest one input, create a documented adapter and report the adapter separately; do not silently substitute a different physical model.

Orbit and force-model cases

Case	Orbit and purpose	Required model	Primary risk tested
A	Molniya-like Earth orbit; baseline HEO propagation	Earth gravity, Sun/Moon third bodies, drag, solar radiation pressure	General HEO accuracy and apogee dwell behavior
B	More extreme eccentricity with low perigee	High-degree gravity, atmosphere, third bodies, radiation pressure	Perigee resolution, step-size control, and numerical stability
C	Geostationary-transfer-like orbit	Gravity, drag near perigee, third bodies, radiation pressure	Strongly nonuniform time scale and rapid perigee dynamics
D	Long-arc unforced/weakly forced case	Gravity plus third bodies; 30, 90, and 365 days	Secular drift and accumulated phase error
E	Maneuver case	Same as A plus impulsive and finite-duration burns near perigee and apogee	Event handling, discontinuities, and post-maneuver recovery
F	Orbit-determination case	Tracking measurements, covariance, and state-transition matrix	Estimation consistency and sensitivity propagation

The Molniya case is mandatory because prior work explicitly evaluates propagation accuracy and computational efficiency for that regime by adapting the geopotential degree and numerical tolerance to the mission error requirement.[4]

Reference truth and comparison hierarchy

No toolkit should be declared "truth" merely because it runs at a smaller step size. Construct a reference solution independently by using at least two numerically distinct high-accuracy configurations: for example, a high-order adaptive integrator with tight tolerances and a second integrator or regularized formulation. Tighten tolerances and increase force-model degree until the difference between the two reference configurations is below the benchmark threshold.

For the most eccentric cases, add a regularized reference using a Sundman or KS formulation. Sundman transformations redistribute computational effort, taking larger effective steps near apogee and smaller ones near perigee.[5] KS scaling has been reported to preserve the Kepler-energy relation with negligible additional cost in the tested formulation.[6] The regularized reference must nevertheless be checked for event-time error, because stopping a propagation at an event can create timing or phasing errors that are not visible in the geometric state error alone.[7]

Metrics

State accuracy

Compute position and velocity errors against the reference at fixed output epochs and at dynamically meaningful events:

- three-dimensional position error;
- velocity error;
- radial, along-track, and cross-track components;
- perigee and apogee epoch error;
- true-anomaly or argument-of-latitude phase error;
- maximum error, root-mean-square error, and 95th-percentile error;
- error growth versus propagation duration.

Report errors both in inertial coordinates and in the local orbital frame. A toolkit can look accurate in Cartesian norm while still accumulating unacceptable along-track timing error.

Force-model and event fidelity

Repeat selected cases while varying gravity degree/order, atmospheric model, third-body inclusion, radiation pressure, maneuver duration, and output cadence. The goal is to separate numerical error from model error. Test event detection at perigee, apogee, ascending node, eclipse entry/exit, ground-station rise/set, and maneuver start/stop. Event accuracy deserves its own score because regularized formulations can lose timing accuracy even when their geometric trajectory is highly accurate.[7]

Runtime and scalability

Measure:

- wall-clock time per propagated day;
- wall-clock time per output state;
- peak memory;
- initialization time;
- scaling with output cadence;
- scaling with force-model complexity;
- scaling from one spacecraft to 10, 100, and 1,000 spacecraft;
- parallel scaling where supported.

Report accuracy at each runtime point so that the output is an accuracy–cost frontier rather than a speed leaderboard.

Robustness

A run passes robustness testing only if it completes without numerical failure and produces finite, physically plausible states. Stress the integrators with tolerance changes, output-cadence changes, event-rich arcs, near-perigee

maneuvers, long propagation durations, and perturbed initial conditions. Repeat selected cases with slightly different integrator settings and check that the scientific conclusion does not change.

Orbit determination and covariance

For the OD case, generate synthetic range, range-rate, and angular measurements from the reference trajectory, add controlled noise, and use identical measurement schedules across toolkits. Compare:

- recovered initial-state error;
- residual RMS and normalized residuals;
- covariance realism using normalized estimation error squared;
- state-transition matrix agreement;
- sensitivity to measurement gaps and perigee-only observations;
- convergence from deliberately biased initial conditions.

This prevents a toolkit from ranking highly on forward propagation while failing in the estimation workflow required for operations.

Acceptance gates

Use gates rather than one composite score:

1. **Numerical validity**: no failed runs, NaNs, unbounded energy error in the unforced test, or unexplained discontinuities.
2. **Accuracy**: meet mission-defined position, velocity, and event-time thresholds for Cases A–F.
3. **Reproducibility**: results remain within a predefined tolerance when output cadence and integrator tolerances are varied within the qualified range.
4. **Operational completeness**: support the required frames, time scales, force models, events, maneuvers, ephemeris export, and covariance/OD functions.
5. **Performance**: meet the target propagation throughput and memory budget at the required accuracy, not at a lower-fidelity setting.

A practical decision rule is to reject any toolkit that fails an accuracy or robustness gate, then choose among the survivors using the accuracy–runtime frontier and operational completeness. Do not compensate for a failed event-time or covariance requirement with faster nominal propagation.

Recommended scorecard

Keep the scorecard visible and multidimensional:

- 30% state and event accuracy;
- 20% long-arc stability;
- 15% force-model completeness;
- 15% orbit-determination and covariance performance;
- 10% runtime and scalability;
- 10% reproducibility, documentation, and deployment friction.

The weights should be changed for the mission. For onboard or near-real-time use, increase runtime and memory. For orbit determination or collision-risk work, increase event, covariance, and state accuracy. For preliminary design, increase throughput and parameter-sweep scalability.

Minimum benchmark deliverables

The benchmark package should contain:

1. version-pinned environment files for Orekit, Nyx, and SatKit;
2. identical input files and a machine-readable case manifest;
3. reference ephemerides and independent-reference convergence evidence;

4. raw state, event, runtime, memory, and OD outputs;
5. automated plots of position, velocity, radial/along-track/cross-track, phase, and event-time errors;
6. an accuracy–runtime Pareto plot for each case;
7. a pass/fail matrix with failure explanations;
8. a reproducibility report showing the effect of tolerance, output cadence, and force-model changes.

Recommended initial implementation order

Start with Cases A–C and the unforced long-arc test. These establish whether each toolkit can reproduce HEO geometry and perigee dynamics before the benchmark becomes entangled with atmosphere, measurement, and maneuver-model differences. Next add the maneuver and event suite, then OD and covariance. Only after those tests pass should the team compare 100- or 1,000-spacecraft scaling.

The most decision-relevant gap is not raw nominal trajectory error; it is the absence of a common, cross-tool HEO validation dataset and a shared force-model configuration. Until that is supplied, claims that one of the three toolkits is "most accurate" should be treated as configuration-specific rather than intrinsic to the software.

References

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